

SRN: PES1UG24AM203

CLASS-AIML D

Lab 1: Requirements Engineering & UML Use-Case Modelling

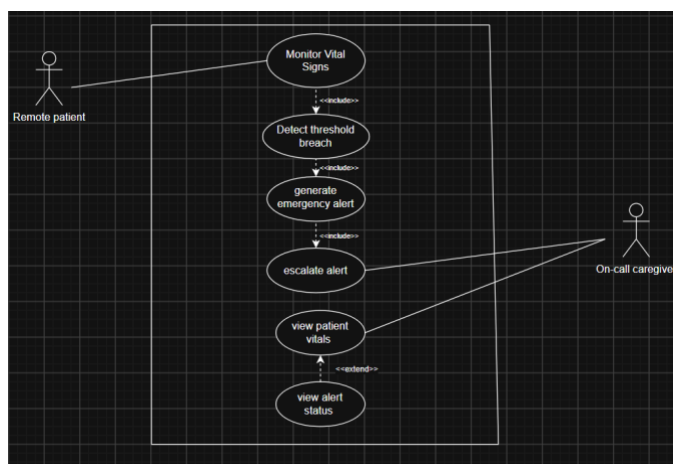
Problem Statement -16: Remote Patient Vitals Alert & Monitoring App

Complete Requirements Table:

Req ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-001	Functional	The system shall continuously monitor patient vital signs including SpO ₂ , heart rate, and blood pressure.	High	Abnormal vital values that exceed configured thresholds are detected within 2 seconds.	Enables early detection of critical conditions
FR-002	Functional	The system shall flag critical vital-sign anomalies and generate an emergency alert.	High	When a configured threshold is breached, an alert is generated for the On-Call Caregiver.	Enables timely response to critical patient conditions.
FR-003	Functional	The system shall escalate emergency alerts to the appropriate On-Call Caregiver based on the configured caregiver matrix.	High	When a critical anomaly is detected, an emergency alert is sent to the assigned On-Call Caregiver	Ensures that the appropriate caregiver is notified promptly during an emergency.
FR-004	Functional	The system shall allow the On-Call Caregiver to view the patient's current vital signs and alert status.	High	The caregiver can view the latest SpO ₂ , heart rate, blood pressure, and active alerts for the patient.	Provides caregivers with the information needed to assess the patient's condition.

FR-005	Functional	The system shall record detected vital-sign anomalies and emergency alerts.	Medium	Each detected anomaly and generated alert is stored with the patient identifier, vital type, recorded value, and timestamp.	Maintains a record of critical events for monitoring and review.
NFR-001	Non-Functional	The system shall support at least 500 concurrent continuous telemetry streams with 99.99% uptime.	High	Performance testing confirms support for at least 500 concurrent telemetry streams while meeting the 99.99% uptime target	Ensures reliable monitoring for a large number of patients.
NFR-002	Non-Functional	The system shall protect patient vital data from unauthorized access during transmission and storage	High	Unauthorized users cannot access patient data, and security testing confirms that patient telemetry is protected during transmission and storage.	Protects sensitive patient information and ensures secure remote monitoring

UML Use-Case Diagram:



Use-Case Flow Specification

Use Case: Generate Emergency Alert

Primary Actor: Remote Patient Monitoring System

Supporting Actor: On-Call Caregiver

Goal: Generate an emergency alert when a patient's vital sign exceeds its configured threshold.

Preconditions

1. Patient telemetry is being received.
2. Vital-sign thresholds are configured.
3. The patient is registered for monitoring.
4. The monitoring system is active.

Postconditions

1. An emergency alert is generated.
2. The alert contains the relevant patient/vital information.
3. The alert is available for escalation to the assigned On-Call Caregiver.
4. The On-Call Caregiver is notified of the emergency alert.

Main Success Scenario

1. The system continuously receives patient vital signs.
2. The system evaluates the vital signs against configured thresholds.
3. The system detects that a vital sign has breached its threshold.
4. The system identifies the condition as a critical anomaly.
5. The system generates an emergency alert.
6. The system forwards the alert for escalation to the assigned On-Call Caregiver.
7. The On-Call Caregiver receives the emergency alert.

Alternate Flow

A1. Vital sign remains within the configured threshold

1. The system receives the patient's vital-sign reading.
2. The system evaluates the reading against the configured threshold.
3. The vital value does not exceed the configured threshold.
4. No emergency alert is generated.
5. The system continues monitoring the patient's vital signs.